Process-Option-Value $(o, (O, E), (V_S, V_V, E_S, E_C))$

if $\exists v \in V_V$: id(v) = hash(o) **then**
  return (V_S, V_V, E_S, E_C)
end

$v \leftarrow$ (hash(o), val(o))
$V_{V'} \leftarrow V_V \cup \{v\}$
let $o' \in O : (o', o) \in E$
let $v' \in V_V$: id(v') = hash(o')
$s' \leftarrow$ spec(v')
if $\exists s \in V_S \mid (s', s) \in E_C \wedge$ name(s) = name(o) **then**
  $E_{S'} \leftarrow E_S \cup \{(v, s)\}$
  $V_{S'} \leftarrow V_S$
  $E_{C'} \leftarrow E_C$
else
  if $o = \text{root}(O, E)$ **then**
    if $V_S = \emptyset$ **then**
      $s \leftarrow$ (name(o), type(val(o)))
      $V_{S'} \leftarrow V_S \cup \{s\}$
    else
      let $s \leftarrow$ root(V_S, E_C)
      $V_{S'} \leftarrow V_S$
    end
    $E_{C'} \leftarrow E_C$
  else
    $s \leftarrow$ (name(o), type(val(o)))
    $V_{S'} \leftarrow V_S \cup \{s\}$
    $E_{C'} \leftarrow E_C \cup \{(s', s)\}$
  end
  $E_{S'} \leftarrow E_S\{(v, s)\}$
end
foreach $o_c \in O \mid (o, o_c) \in E$ **do**
  $(V_{S'}, V_{V'}, E_{S'}, E_{C'}) \leftarrow$ Process-Option-Value $(o_c, (O, E), (V_{S'}, V_{V'}, E_{S'}, E_{C'}))$
  let $v_c \in V_{V'} \mid$ id(v_c) = hash(o_c)
  $E_{C'} \leftarrow E_C \cup \{(v, v_c)\}$
end
return $(V_{S'}, V_{V'}, E_{S'}, E_{C'})$

Process-Option-Value $(o, (O, E), (V_S, V_V, E_S, E_C))$

if $\exists v \in V_V : \mathrm{id}(v) = \mathrm{hash}(o)$ **then**
    return (V_S, V_V, E_S, E_C)
end

$v \leftarrow (\mathrm{hash}(o), \mathrm{val}(o))$
$V_{V'} \leftarrow V_V \cup \{v\}$
if $V_S \neq \emptyset \wedge o = \mathrm{root}(O, E)$ **then**
    let $s \leftarrow \mathrm{root}(V_S, E_C)$
    $E_{S'} \leftarrow E_S\{(v, s)\}$
else
    let $o' \in O : (o', o) \in E$
    let $v' \in V_V : \mathrm{id}(v') = \mathrm{hash}(o')$
    $s' \leftarrow \mathrm{spec}(v')$
    if $\exists s \in V_S \mid (s', s) \in E_C \wedge \mathrm{name}(s) = \mathrm{name}(o)$ **then**
        $E_{S'} \leftarrow E_S \cup \{(v, s)\}$
        $V_{S'} \leftarrow V_S$
        $E_{C'} \leftarrow E_C$
    else
        $s \leftarrow (\mathrm{name}(o), \mathrm{type}(\mathrm{val}(o)))$
        $E_{S'} \leftarrow E_S\{(v, s)\}$
        $V_{S'} \leftarrow V_S \cup \{s\}$
        if $o \neq \mathrm{root}(O, E)$ **then**
            $E_{C'} \leftarrow E_C \cup \{(s', s)\}$
        else
            $E_{C'} \leftarrow E_C$
        end
    end
end
foreach $o_c \in O \mid (o, o_c) \in E$ **do**
    $(V_{S'}, V_{V'}, E_{S'}, E_{C'}) \leftarrow$ Process-Option-Value $(o_c, (O, E), (V_{S'}, V_{V'}, E_{S'}, E_{C'}))$
    let $v_c \in V_{V'} \mid \mathrm{id}(v_c) = \mathrm{hash}(o_c)$
    $E_{C'} \leftarrow E_C \cup \{(v, v_c)\}$
end
return $(V_{S'}, V_{V'}, E_{S'}, E_{C'})$